Linearising Maier-Saupe

Complex disordered systems: Liquid crystals

In the Maier-Saupe mean-field theory of nematic liquid crystals, the orientation distribution function of rod-like molecules is given by

$$p(\theta) = \frac{1}{Z}\exp\left(\lambda P_2(\cos\theta)\right), \qquad Z = \int_0^\pi \exp\left(\lambda P_2(\cos\theta)\right)\sin\theta\, d\theta,$$

where $\lambda = uS/(k_BT)$ and

$$S = \langle P_2(\cos\theta)\rangle = \frac{\int_0^\pi P_2(\cos\theta)p(\theta)\sin\theta\, d\theta}{\int_0^\pi p(\theta)\sin\theta\, d\theta}.$$

This leads to the implicit ("self-consistent") relation

$$S = \frac{\int_0^\pi P_2(\cos\theta)e^{\lambda P_2(\cos\theta)}\sin\theta\, d\theta}{\int_0^\pi e^{\lambda P_2(\cos\theta)}\sin\theta\, d\theta}.$$

You may use

$$\int_0^\pi P_2(\cos\theta)\sin\theta\, d\theta = 0, \qquad \int_0^\pi [P_2(\cos\theta)]^2\sin\theta\, d\theta = \frac{2}{5}.$$

1. Explain why the above equation is implicit and what it means to linearise it near the isotropic phase.
2. Assuming λ is small, expand
$$e^{\lambda P_2(\cos\theta)}$$
to first order in λ.
3. Using your expansion derive the linearised relation between S and λ.
4. Using $\lambda = uS/(k_BT)$, determine the value of k_BT/u at which a non-zero solution for S first appears.

Solution

1. **Why the equation is implicit and what linearization means**

The equation is implicit because S appears on both sides: explicitly on the left side, and implicitly on the right side through $\lambda = uS/(k_BT)$. To solve for S, we need to find values that satisfy this self-consistency condition.

Linearization near the isotropic phase means we expand around $S = 0$ (the isotropic state where there's no preferred orientation) and keep only first-order terms in small parameters. This allows us to find the critical temperature where nematic order first appears.

2. **Expand the exponential to first order**

For small λ, we expand:

$$e^{\lambda P_2(\cos\theta)} \approx 1 + \lambda P_2(\cos\theta)$$

3. **Derive the linearized relation**

Substituting the expansion into the self-consistent equation:

Numerator:

$$\int_0^\pi P_2(\cos\theta)[1+\lambda P_2(\cos\theta)]\sin\theta\,d\theta = \int_0^\pi P_2(\cos\theta)\sin\theta\,d\theta + \lambda\int_0^\pi [P_2(\cos\theta)]^2\sin\theta\,d\theta$$

Using the given integrals:

$$= 0 + \lambda\cdot\frac{2}{5} = \frac{2\lambda}{5}$$

Denominator:

$$\int_0^\pi [1+\lambda P_2(\cos\theta)]\sin\theta\,d\theta = \int_0^\pi \sin\theta\,d\theta + \lambda\int_0^\pi P_2(\cos\theta)\sin\theta\,d\theta$$

$$= 2 + \lambda\cdot 0 = 2$$

Therefore, the linearized relation is:

$$S = \frac{2\lambda/5}{2} = \frac{\lambda}{5}$$

4. **Find the critical temperature**

Substituting $\lambda = uS/(k_BT)$:

$$S = \frac{uS}{5k_BT}$$

For a non-trivial solution ($S \neq 0$), we can divide both sides by S:

$$1 = \frac{u}{5k_BT}$$

Therefore:

$$\frac{k_BT}{u} = \frac{1}{5}$$

The critical value is $k_BT_c/u = 1/5$, which is the point at which the instability to the order parameter fluctuations sets in and corresponds therefore to the spinodal point of the isotropic phase.

Let's think for a moment however about what we have done: we have assumed to stay in the isotropic phase and looked at small perturbations around it that satisfy the self-consistent equation. We have found that trivially the isotropic phase is always a solution. However we have also found that at $k_BT/u = 1/5$ we can change any value of S (provided that it is small enough for the linearisation to hold) into a solution of the self-consistent equation.

This means that at such temperature the isotropic phase becomes marginally stable: any small perturbation in S will not be restored back to zero. This is the definition of a spinodal point.